From prediction error to conscious access

01 Reference errors

Separate causal reference windows for each channel

$z = (\varepsilon - \mu_{ref})/\sigma_{ref}$

02 Effective gains

ACh → exteroceptive gain
NE + somatic marker → interoceptive gain

Clamp, then normalize

03 Excess surprise

$|z|_{ex} = |z| - \sqrt{2/\pi}$

Relative effective gains weight channel-specific excess magnitudes

04 Accumulation

$S_{inst} = \sum_j \Pi^{eff}_{rel,j} |z^{eff}_j|_{ex}$

Leaky integration → S_t
Signal may be negative

05 Ignition

$S_t > \theta_{eff,t}$

Global workspace broadcast; partial reset and refractory dynamics

Somatic-marker pathway

$\Pi^{eff}_i = g_{NE} \Pi_i \exp(\beta_{SM} \hat{M}(c, a))$ (before clamping)

Allostatic control

Cost raises θ; expected value lowers θ.

Mathematical Specification §§2–4, 13; normalization follows clamping. Default untransformed error channels shown.